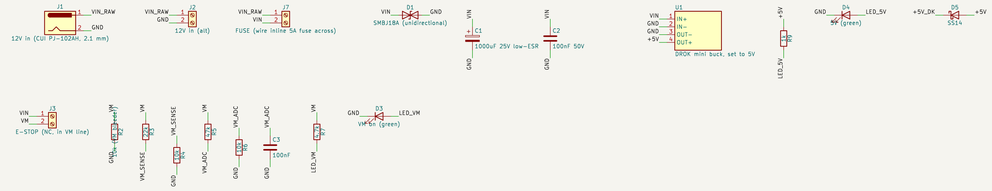
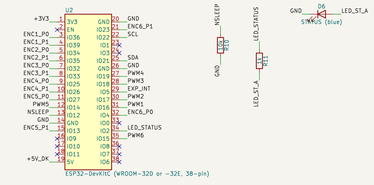
POWER IN, REVERSE PROTECTION, E-STOP (no relay: NC contacts in VM line)

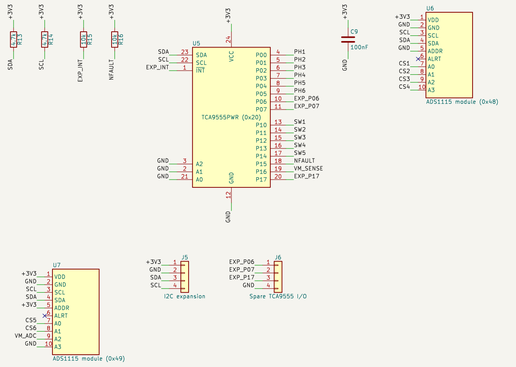


5 V: DROK buck module in 4-pin socket (IN+ IN- OUT- OUT+) + DEVKIT 5V FEED

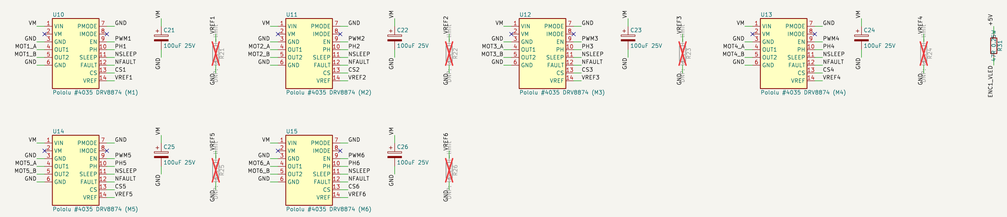
ESP32-DevKitC, WROOM-32D or -32E (plug-in, 2x 1x19 female headers)



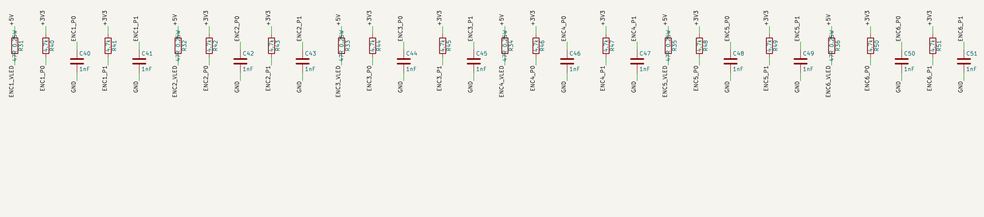
I2C: TCA9555 (0x20) + 2x ADS1115 modules (0x48, 0x49)



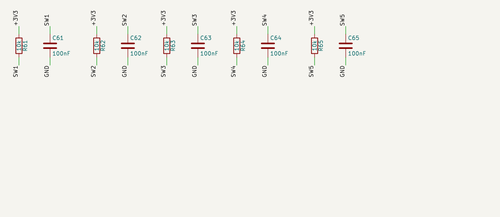
MOTOR DRIVERS: 6x Pololu DRV8874 carrier (plug-in) — PIN ORDER PLACEHOLDER



ENCODER FRONT END: 47R VLED from 5V, 4.7k pull-ups to 3.3V, 1nF, direct to ESP32



HOME SWITCHES (to TCA9555 P1.0–P1.4)



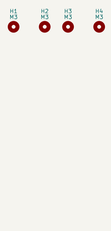
ROBOT CABLE: DD-50 female (SCORBOT-ER 4u manual Rev E)



POWER FLAGS



MOUNTING



NOTES

1. Pololu DRV8874 carrier pin ORDER is a placeholder (names are correct). Fix design.CARRIER_ROWS and regenerate.
2. DB50 pinout from SCORBOT-ER 4u User Manual ch. 8 (same as ER III Table D-1). Pins 37–44 unused, 22 = gripper switch (no connection in arm).
3. Buck socket follows the DROK module (EN unused). ADS1115 socket follows Adafruit #1085; other boards differ.
4. Carrier current limit with SLEEP at 3.3 V is about 2.9 A (VREF pulled to SLEEP by 10k on the carrier, CS has 2.49k). R21–R26 (DNP) lower it.
5. Current sense: CS reads 1.12 V/A into ADS1115 modules (use the +/−4.096 V range). VM monitored via 47k/10k on ADS1115 #2 A2.

Pololu carrier pin ORDER is a placeholder — see README
Generated by generator/gen_schematic.py from design.py

Sheet: /
File: scorbot_controller.kicad_sch

Title: Scorbot ER-4u ESP32 master board

Size: A1	Date: 2026-10-09	Rev: 0.1 (generated)
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/1